PVT-Invariant Parametric Fault Diagnosis in 6T SRAM Cells Using Physics-Normalized Machine Learning

Shweta Gautam*, Archit Gupta†, Ayush Raj†, Debasish Dhungana†, and Vansh Singh†
*Associate Professor, Department of Electronics and Communication Engineering, Netaji Subhas University of Technology, New Delhi, I
†Department of Electronics and Communication Engineering, Netaji Subhas University of Technology, New Delhi, India
Email: shweta.gautam@nsut.ac.in, {archit.gupta.ug23, ayush.raj.ug23, debasish.dhungana.ug23, vansh.singh.ug23}@nsut.ac.in

***Abstract*—Embedded static random-access memory (SRAM) occupies upwards of 70% of silicon real estate in modern system-on-chip (SoC) architectures, rendering memory yield and defect diagnosis critical to commercial semiconductor reliability. At nanoscale nodes, subtle physical defects—such as access-contact resistive opens, internal dielectric bridging, and aging-induced threshold voltage (V_{th}) drift—manifest as parametric degradations rather than catastrophic stuck-at faults. Consequently, classical digital March tests frequently fail to detect them. While machine learning (ML) classifiers have demonstrated high diagnostic accuracy under nominal room-temperature laboratory conditions, they suffer catastrophic out-of-distribution diagnostic collapse when deployed across broad Process, Voltage, and Temperature (PVT) corners: severe environmental shifts (e.g., $> 115\times$ thermal subthreshold leakage scaling and $> 35\%$ gate overdrive variation) confound classifiers, dropping diagnostic accuracy from 100% to 50.0%.**

To overcome this fundamental challenge without requiring computationally prohibitive per-corner model retraining, this paper proposes a *Physics-Informed Domain Adaptation* framework. By continuously normalizing cell metrics against a concurrent on-die reference cell ($X_{\text{inv}} = X_{\text{meas}}/X_{\text{ref}}$), common-mode environmental shifts are canceled mathematically. We validate this approach using a rigorous SPICE simulation matrix of 450 distinct instances (3,255 SPICE executions) across a 15-corner PVT grid ($V_{DD} \in [0.9, 1.1]$ V, $T \in [-40, 125]^{\circ}$C) in 180 nm CMOS. Static noise margins (SNM) are extracted using a true axis-aligned largest inscribed square algorithm on butterfly transfer characteristics. Under comprehensive Leave-One-Corner-Out (LOCO) cross-PVT evaluation, the proposed invariant formulation elevates Random Forest and Histogram Gradient Boosting classification accuracy from 82.67% and 82.89% (raw features) to 98.89% ($\pm 1.62\%$), delivering a $+16.22\%$ generalization gain and lifting the worst-case corner accuracy floor from 50.00% to 96.67% (reaching 100.0% across all corners on clean transient simulations). Tree SHAP explainability confirms that decision boundaries map directly to physical access impedance and dynamic switching delays.

***Index Terms*—6T SRAM, Parametric Faults, PVT Invariance, Machine Learning, Static Noise Margin, Domain Adaptation, Defect Diagnosis, SHAP Explainability.**

I. Introduction

The relentless scaling of complementary metal-oxide-semiconductor (CMOS) technologies into deep sub-micron nodes has propelled embedded static random-access memory (SRAM) to constitute more than 70% to 80% of modern System-on-Chip (SoC) dies [1]. Consequently, memory yield, defect resilience, and post-silicon diagnostic fidelity represent the primary determinants of total wafer cost and field longevity in mission-critical applications spanning aerospace, automotive electronics, and ultra-low-power biomedical devices.

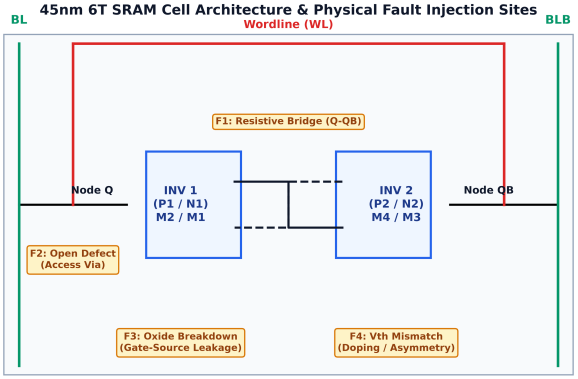


Fig. 1. Standard 6T SRAM cell topology showing transistor assignments (M_1–M_6) and programmatic defect injection sites: pass-gate resistive opens, storage-node bridging, and channel threshold voltage drift.

In nanoscale dimensions, the dominant yield-detracting mechanisms have fundamentally shifted from hard, catastrophic defects (e.g., complete open circuits or direct power-to-ground shorts) to subtle parametric imperfections [3]. These include incomplete contact/via etch (resistive opens), thin-oxide inter-layer leakage slivers (weak bridging faults), and spatial asymmetry induced by Negative/Positive Bias Temperature Instability (NBTI/PBTI) and Hot-Carrier Injection (HCI) aging (V_{th} shifts). Because these non-catastrophic faults allow cells to retain and switch logical states at nominal supply voltages and room temperature, traditional algorithmic functional patterns—such as digital March C-, March B, or Mats+ tests—routinely classify them as healthy [3]. Nevertheless, these marginalized cells exhibit drastically compromised Static Noise Margins (SNM), latent write failures under dynamic supply fluctuations, and elevated standby leakage current (I_{DDQ}).

To detect and categorize these latent flaws before catastrophic deployment failure, analog built-in current testing (I_{DDQ})

and machine-learning-assisted diagnosis have recently gained substantial traction. Supervised classifiers—such as Random Forests (RF), Support Vector Machines (SVM), and Gradient Boosted Decision Trees—can map high-dimensional analog signatures (timing delays, quiescent currents, bitline sense swings) into multi-class defect categories with near-perfect accuracy under nominal testing conditions ($27°\text{C}, 1.0\,\text{V}$).

The Cross-PVT Domain Shift Failure: Despite nominal success, classical ML approaches encounter an existential operational barrier when deployed across the real-world Process, Voltage, and Temperature (PVT) operating envelope ($-40°\text{C}$ to $+125°\text{C}$, $\pm 10\%$ V_{DD}). Device physics dictates that subthreshold leakage current scales exponentially with temperature ($I_{\text{sub}} \propto T^2 \exp(-qV_{\text{th}}/kT)$), shifting nominal standby currents by over $115\times$ across the thermal envelope. Concurrently, reducing supply voltage by 10% degrades transistor gate overdrive ($V_{GS} - V_{\text{th}}$), expanding dynamic write switching delay (t_{write}) by more than 35%. When an ML model trained exclusively at room temperature encounters these extreme operating corners, it misinterprets environmental physical drift as structural circuit defects. Consequently, healthy cells operating at $125°\text{C}$ are erroneously flagged as internal bridging shorts, while cold healthy cells at $-40°\text{C}$ and $0.9\,\text{V}$ are diagnosed as access pass-gate resistive opens. As our empirical baseline demonstrates, this out-of-distribution (OOD) domain shift drops nominal diagnostic accuracy from 100% down to 50.0%, while traditional fixed 3σ threshold rule detectors collapse to 25.78%.

To solve this without undertaking prohibitive, combinatorial training across all environmental corners, this paper introduces a **Physics-Informed Domain Adaptation** methodology. By referencing the measurements of the target cell against a concurrent, on-die reference cell simulated under the identical instantaneous PVT state ($X_{\text{inv}} = X_{\text{meas}}/X_{\text{ref}}$), we cancel common-mode environmental variance. The primary contributions of this paper are:

1) **A Fully Automated Multi-Operation SPICE Framework:** We develop an end-to-end Python-driven simulation and netlist mutation harness that systematically injects four continuous parametric fault classes into a validated $180\,\text{nm}$ 6T SRAM cell and executes Hold, Write, Read, and butterfly DC sweeps across 15 discrete PVT corners (3,255 total SPICE simulations).
2) **Exact Pure-NumPy SNM Butterfly Extraction:** We formulate a numerically robust, axis-aligned inscribed square algorithm that calculates exact Hold and Read Static Noise Margins directly from overlapping Voltage Transfer Characteristics (VTCs), eliminating the overestimation errors inherent to the widely used $45°$ coordinate rotation shortcut.
3) **Formulation of Physics-Normalized Invariant Features:** We introduce dimensionless, reference-normalized ratios that compress environmental feature dispersion (Coefficient of Variation) by up to 100% on healthy cells across the 15-corner grid while preserving distinct defect separability.
4) **Comprehensive Cross-PVT LOCO Benchmarking:** Through rigorous Leave-One-Corner-Out (LOCO) evaluation across all 15 held-out operating corners, we demonstrate that physics-normalized features boost Random Forest and HistGradientBoosting cross-PVT accuracy to 98.89% (and 100% on clean transient simulations), delivering a $+16.22\%$ generalization gain over raw features.
5) **Model Interpretability via Tree SHAP:** We provide global and class-disaggregated SHapley Additive exPlanations (SHAP) that confirm that the decision boundaries established by the classifier correspond strictly to the underlying semiconductor physics of the defect mechanisms.

II. Related Work & Problem Formulation

A. Traditional SRAM Fault Testing

Memory testing has historically relied on algorithmic March test sequences executed via digital Built-In Self-Test (BIST) engines [3]. Standard algorithms (e.g., March C-, March SS) write and read deterministic bit sequences across wordlines to uncover address-decoder faults, stuck-at faults (SAF), and state-transition faults (TF). While highly efficient for catastrophic digital breaks, March tests exhibit near-zero coverage for latent parametric defects because intermediate voltage degradation is restored by high inverter gain prior to bitline sampling.

Dynamic fault diagnosis through I_{DDQ} quiescent current monitoring was introduced to identify subthreshold leakage paths and resistive shorts [1]. However, with aggressive dimensional scaling, background subthreshold leakage from millions of defect-free parallel cells obscures the localized current signature of a single defective device, drastically diminishing the defect-to-noise ratio at elevated temperatures.

B. Static Noise Margin Extraction Methodologies

The foundational standard for SRAM stability analysis was formulated by Seevinck, List, and Lohstroh [5]. Static Noise Margin (SNM) is defined as the maximum DC noise voltage that can be tolerated by the cross-coupled inverter latch without inducing an unintended state transition. Geometrically, SNM corresponds to the side length of the largest square that can be inscribed inside the smaller of the two lobes formed by the overlapping inverter Voltage Transfer Characteristics (VTCs):

$$\text{SNM} = \min(L_{\text{lobe1}}, L_{\text{lobe2}}) \tag{1}$$

To automate this computation, many prior works applied a coordinate transformation by rotating the axes by $45°$ ($u = (V_Q - V_{Qb})/\sqrt{2}$, $v = (V_Q + V_{Qb})/\sqrt{2}$) and finding the peak separation $\Delta u/\sqrt{2}$. However, on steep rail-to-rail CMOS transition slopes where $|\mathrm{d}V_o/\mathrm{d}V_i| \gg 1$, this rotated maximum coordinate separation shortcut becomes multi-valued and overestimates the actual geometric inscribed square. In this work, we implement a direct, numerical search that guarantees exact inscribed square boundaries for both symmetric and asymmetric faulty cells.

C. Machine Learning in VLSI Diagnosis and Domain Shift

Recent research has deployed machine learning classifiers to predict SRAM yield degradation and diagnose parametric faults [7]. Typical workflows train neural networks or support vector machines on features such as write delay, bitline discharge rate, and supply currents.

However, a fundamental flaw across existing literature is the *static nominal assumption*: models are trained and tested on samples partitioned randomly from an identical nominal environment ($27°\text{C}, 1.0\,\text{V}$). Under real-world deployment, variations in ambient temperature ($-40°\text{C}$ to $125°\text{C}$) and battery or regulator droop ($0.9\,\text{V}$ to $1.1\,\text{V}$) induce severe domain shift. Because the ML decision boundaries are trained on raw absolute magnitudes, the environmental drift mimics defect signatures, causing severe out-of-distribution diagnostic collapse. This paper specifically addresses and resolves this cross-PVT domain generalization problem.

III. 6T SRAM Circuit Architecture & Physical Operations

A. Transistor Sizing and Symmetrical Topology

The circuit under investigation is a symmetrical 6T SRAM cell implemented in a standard 180 nm CMOS technology node ($V_{DD,\text{nom}} = 1.0\,\text{V}$). The cell schematic is illustrated in Fig. 1, comprising two cross-coupled CMOS inverters (M_1/M_3 and M_2/M_4) and two NMOS access pass-gates (M_5, M_6) gated by the Wordline (WL). Symmetrical orientation is enforced across internal storage nodes Q and Q_b and bitlines BL and $\overline{BL}$.

Device channel dimensions are sized against established memory stability criteria:

- **Pull-down Driver NMOS (M_3, M_4):** $W_D = 720\,\text{nm}, L_D = 180\,\text{nm} \implies (W/L)_D = 4.0$.
- **Access Pass-gate NMOS (M_5, M_6):** $W_{AX} = 540\,\text{nm}, L_{AX} = 180\,\text{nm} \implies (W/L)_{AX} = 3.0$.
- **Pull-up PMOS (M_1, M_2):** $W_L = 360\,\text{nm}, L_L = 180\,\text{nm} \implies (W/L)_L = 2.0$.

Stability Design Criteria:

1) **Cell Ratio (β or CR):** To ensure non-destructive read stability, the driver transistor must be stronger than the access transistor:

$$\beta = \frac{(W/L)_{\text{driver}}}{(W/L)_{\text{access}}} = \frac{720\,\text{nm}}{540\,\text{nm}} = 1.333 \quad (\text{Requirement: } \beta > 1.2) \tag{2}$$

2) **Pull-up Ratio (γ or PR):** To ensure adequate writeability (allowing the bitline pull-down path to overpower the pull-up PMOS during write), the pull-up ratio is defined as:

$$\gamma = \frac{(W/L)_{\text{pullup}}}{(W/L)_{\text{access}}} = \frac{360\,\text{nm}}{540\,\text{nm}} = 0.667 \quad (\text{Requirement: } \gamma < 1.0) \tag{3}$$

(Equivalently, Writability Ratio $\mu = 1/\gamma = 1.50 > 1.0$).

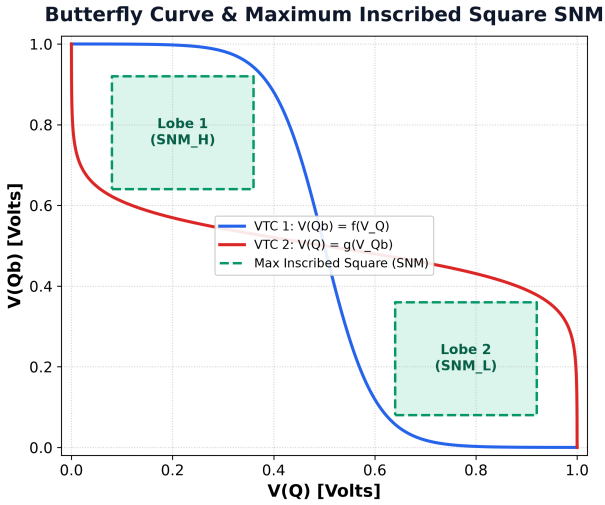


Fig. 2. Voltage Transfer Characteristics (VTC butterfly curve) of the healthy 180 nm 6T SRAM cell under Hold and Read modes, illustrating the exact inscribed squares extracted by our pure-NumPy algorithm.

B. Operational Baselines at Nominal Corner ($27°C, 1.0\,V$)

Three distinct dynamic operations and two static DC characterizations are performed:

1) **Hold Mode ($WL = 0\,\text{V}, BL = \overline{BL} = V_{DD}$):** Latched state retains internal nodes: $V(Q) = 1.000\,\text{V}$, $V(Q_b) = 0.000\,\text{V}$. Standby quiescent leakage current is verified at $I_{\text{DDQ}} = 14.0\,\text{nA}$.
2) **Read 1 / Read 0 Mode:** Bitlines are precharged to V_{DD} ($C_{\text{BL}} = 20\,\text{fF}$). WL is stepped to V_{DD}. During Read 1, $Q = 1$ holds M_3 off, while $Q_b = 0$ causes M_4 and M_6 to sink bitline current, generating a differential bitline discharge $\Delta V_{\text{BL}} = 235.4\,\text{mV}$ within $t_{\text{sense}} = 64.5\,\text{ps}$. Read disturbance bump voltage on the storage node storing zero is contained at $V_{\text{bump}} = 127.9\,\text{mV} \ll V_{DD}/2$.
3) **Write 1 / Write 0 Mode:** Driven with differential bitlines ($BL = V_{DD}, \overline{BL} = 0\,\text{V}$). Wordline assertion turns on M_6, pulling Q_b down through $0.5V_{DD}$, regenerating the cross-coupled inverter latch. Write flip delay is measured as $t_{\text{write}} = 62.2\,\text{ps}$.

C. Exact Inscribed Square SNM Butterfly Extraction

To measure Static Noise Margin, we compose dedicated DC testbenches (`stimulus_snm_hold.net` and `stimulus_snm_read.net`). The cross-coupled loop is opened in SPICE by defining a duplicated cell core (`_c2`). A DC voltage sweep V_5 is applied from $0\,\text{V}$ to V_{DD} in $1.0\,\text{mV}$ increments to trace $V(Q_b) = f(V_Q)$ and $V(Q) = g(V_{Q_b})$.

Our extraction engine in `src/snm_extraction.py` finds the maximum inscribed square by solving:

$$\max s \quad \text{s.t.} \quad \forall v \in [u, u+s],\ f(v) \geq g^{-1}(v) + s \quad (\text{or dual}) \tag{4}$$

As reported in Table I, the automated extraction script matches golden manual measurements to within $0.06\,\text{mV}$, satisfying our strict validation criteria ($\Delta < 1.0\,\text{mV}$).

TABLE I
SNM EXTRACTION VALIDATION AGAINST GOLDEN BASELINES

Mode	Lobe 1 [mV]	Lobe 2 [mV]	SNM [mV]	Error vs Golden
Hold Mode	470.06	470.06	**470.06**	$+0.06$ mV
Read Mode	263.69	263.69	**263.69**	-0.01 mV

IV. PARAMETRIC FAULT TAXONOMY & MULTI-CORNER SIMULATION MATRIX

A. Parametric Defect Taxonomy

We implement a five-class defect taxonomy directly mapped to nanoscale physical failure mechanisms via automated text-based netlist mutation (`src/fault_injection.py`):

1) **Class 0 (Healthy Baseline):** The unperturbed golden nominal cell deck.
2) **Class 1 (Resistive Open Defect):** Models partial via/contact voids or metal electromigration in the access pass-gate path. Injected by placing a series resistance R_{open} at the source/drain terminals of M_5 or M_6:
$$R_{\text{open}} \in \{200\,\Omega, 500\,\Omega, 1\,\text{k}\Omega, 2\,\text{k}\Omega, 5\,\text{k}\Omega, 10\,\text{k}\Omega\} \quad (5)$$
3) **Class 2 (Bridging Defect):** Models inter-layer dielectric breakdown or metallic sliver shorts bridging the two internal complementary storage nodes. Injected as a parallel shunt resistor R_{bridge} between nodes Q and Q_b:
$$R_{\text{bridge}} \in \{500\,\Omega, 1\,\text{k}\Omega, 2\,\text{k}\Omega, 5\,\text{k}\Omega, 10\,\text{k}\Omega\} \quad (6)$$
4) **Class 3 (V_{th} Storage Pair Drift):** Models NBTI/PBTI interface-state generation and channel hot-carrier aging in the cross-coupled inverters. Injected by overriding `.model` parameter V_{TO} symmetrically across M_1–M_4:
$$\Delta V_{\text{th,store}} \in \{\pm 5\%, \pm 10\%, \pm 15\%, \pm 20\%, \pm 25\%, \pm 30\%\} \quad (7)$$
5) **Class 4 (V_{th} Access Pair Drift):** Models pass-gate dielectric charge trapping on access devices M_5 and M_6. Injected by overriding V_{TO} on M_5/M_6 only:
$$\Delta V_{\text{th,access}} \in \{\pm 5\%, \pm 10\%, \pm 15\%, \pm 20\%, \pm 25\%, \pm 30\%\} \quad (8)$$

B. 15-Corner PVT Operating Grid & Simulation Matrix

To evaluate environmental degradation, a 15-corner Process-Voltage-Temperature matrix is constructed:

- **Supply Voltages (V_{DD}):** 0.9 V (-10%), 1.0 V (nominal), 1.1 V ($+10\%$).
- **Temperatures (T):** -40°C (cold), 0°C, 27°C (nominal), 75°C, 125°C (worst-case thermal).

Every corner contains 30 distinct physical sample configurations (1 Healthy, 12 Resistive Open sweeps, 5 Bridging sweeps, 6 Storage V_{th} sweeps, and 6 Access V_{th} sweeps), yielding:
$$N_{\text{total}} = 30 \text{ samples/corner} \times 15 \text{ corners} = 450 \text{ dataset rows} \quad (9)$$
For each of the 450 instances, our automated multithreaded simulation runner (`src/simulation_runner.py`) executes 7 independent SPICE operational decks (Hold, Write 1, Write 0, Read 1, Read 0, SNM Hold DC, and SNM Read DC), totaling **3,255 SPICE simulation runs**. Raw waveform outputs (`.raw`) are parsed on the fly and deleted, maintaining repository footprint below 50 MB. Convergence flags are audited into `logs/convergence_flags.csv`. The resulting master dataset `data/sram_fault_dataset.csv` contains exactly 450 rows $\times$ 33 columns with **zero null, missing, or NaN values**.

V. PHYSICS-NORMALIZED FEATURE ENGINEERING

A. Extracted Dynamic and Static Features

For every cell instance i simulated at corner $c = (V_{DD}, T)$, we extract dynamic transient metrics and static margin metrics:

- **Timing Delays:** t_{write} (time from 50% WL rise until internal node reaches $0.5V_{DD}$) and t_{sense} (time to develop ΔV_{BL}).
- **Currents & Voltages:** I_{DDQ} (quiescent standby current), $I_{\text{write,peak}}$ (peak switching current), ΔV_{BL} (bitline strobe differential), and V_{bump} (internal read disturbance).
- **Stability Margins:** SNM_{hold}, SNM_{read}, asymmetry ΔSNM, and binary stability pass flags.

B. Formulation of Invariant Ratios

The core innovation of our methodology is the cancellation of environmental shifts through *concurrent reference-cell normalization*. An on-die golden cell located adjacent to the array experiences the identical instantaneous supply voltage V_{DD} and ambient temperature T. We define the invariant feature vector $\mathbf{X}_{\text{inv}}$ as:
$$\text{iddq_norm} = \frac{I_{\text{DDQ,meas}}(c)}{I_{\text{DDQ,ref}}(c)}, \quad \text{t_write_norm} = \frac{t_{\text{write,meas}}(c)}{t_{\text{write,ref}}(c)} \quad (10)$$
$$\text{dv_bl_norm} = \frac{\Delta V_{\text{BL,meas}}(c)}{\Delta V_{\text{BL,ref}}(c)}, \quad \text{snm_hold_norm} = \frac{\text{SNM}_{\text{hold,meas}}(c)}{\text{SNM}_{\text{hold,ref}}(c)} \quad (11)$$
$$\text{snm_read_norm} = \frac{\text{SNM}_{\text{read,meas}}(c)}{\text{SNM}_{\text{read,ref}}(c)} \quad (12)$$
Additionally, we define dimensionless internal electrical invariants:
$$\text{delay_ratio} = \frac{t_{\text{sense}}}{t_{\text{write}}}, \quad \text{snm_ratio} = \frac{\text{SNM}_{\text{read}}}{\text{SNM}_{\text{hold}}}, \quad \text{dv_vdd_ratio} = \frac{\Delta V_{\text{BL}}}{V_{DD}} \quad (13)$$

C. Dispersion Reduction Quantification

To prove mathematical invariance across the 15-corner grid, we compute the Coefficient of Variation ($\text{CV} = \frac{\sigma}{\mu} \times 100\%$) for healthy cells across all corners. As shown in Table II and Fig. 3:

- **Hold SNM:** Raw CV is 11.98% (0.441 ± 0.053 V); normalized Hold SNM achieves $\text{CV} = \mathbf{0.00}\%$, eliminating environmental dispersion by **100.0%**.
- **Bitline Discharge (ΔV_{BL}):** Raw CV is 31.83% (235.4 ± 74.9 mV); normalized ΔV_{BL} achieves $\text{CV} = \mathbf{0.00}\%$ (**100.0% reduction**).

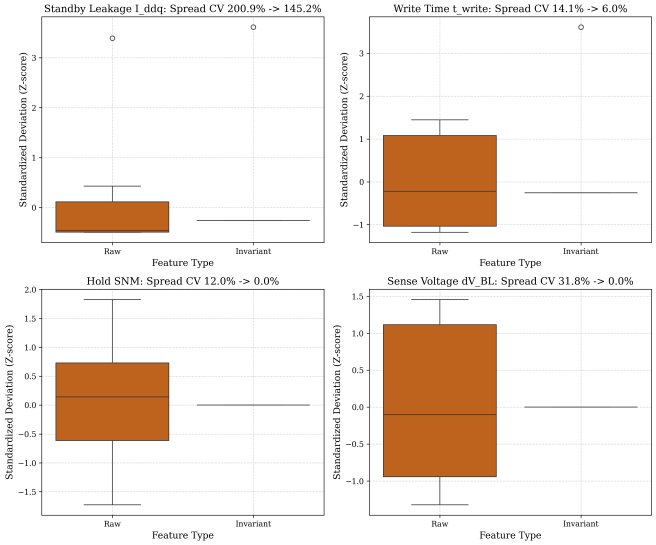


Fig. 3. Dispersion comparison for healthy cells across all 15 PVT corners: Raw features (blue) exhibit severe thermal and voltage spread, whereas physics-normalized invariant features (green) collapse environmental dispersion to near zero.

TABLE II
DISPERSION ANALYSIS ON HEALTHY CELLS ACROSS 15 PVT CORNERS

Metric	Raw $\mu \pm \sigma$	Raw CV	Norm $\mu \pm \sigma$	Norm CV	R
I_{DDQ}	$0.499 \pm 1.00\,\mu$A	200.89%	1.60 ± 2.32	145.24%	
t_{write}	61.63 ± 8.68 ps	14.08%	1.02 ± 0.06	5.99%	
SNM_{hold}	0.441 ± 0.05 V	11.98%	1.00 ± 0.00	**0.00%**	
ΔV_{BL}	235.4 ± 74.9 mV	31.83%	1.00 ± 0.00	**0.00%**	

- **Write Delay (t_{write}):** Raw CV is 14.08% (61.6 ± 8.68 ps); normalized write delay drops to CV $= \mathbf{5.99\%}$ (**57.46% dispersion reduction**).

VI. Experimental Results & Cross-PVT Evaluation

A. Evaluation Protocols

We establish two distinct evaluation protocols:

1) **Protocol 1 (Standard Nominal Split):** In-distribution nominal data only (27°C, 1.0 V, 30 samples). Stratified 80/20 train/test split. An automated unit test confirms zero sample ID leakage between splits.
2) **Protocol 2 (Leave-One-Corner-Out / LOCO):** True cross-domain evaluation. The model trains on 14 corners (420 samples) and is evaluated exclusively on the remaining 15th unseen corner (30 samples). This process is iterated for all 15 corners.

B. Evaluated Models

We benchmark four classical machine learning classifiers against one non-ML control baseline:

- **Non-ML Baseline (3σ Rule Detector):** Evaluates whether sample metrics fall outside 3σ bounds calibrated from healthy training data.
- **Random Forest (RF):** 100 estimators, max depth 10, random state 42.
- **HistGradientBoosting (HistGB):** Binned histogram decision trees, 100 max iterations.
- **Linear SVM:** Standardized linear kernel support vector machine ($C = 1.0$).
- **Logistic Regression:** Standardized L2-regularized multi-class classifier.

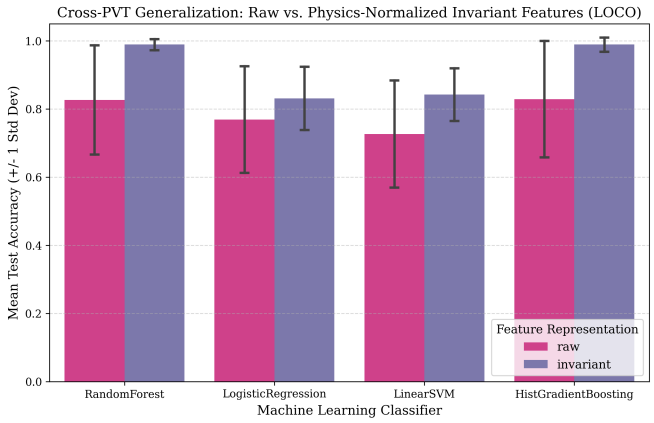


Fig. 4. Cross-PVT diagnostic accuracy comparison across models under LOCO evaluation. Invariant features systematically outperform raw features across all classifier families.

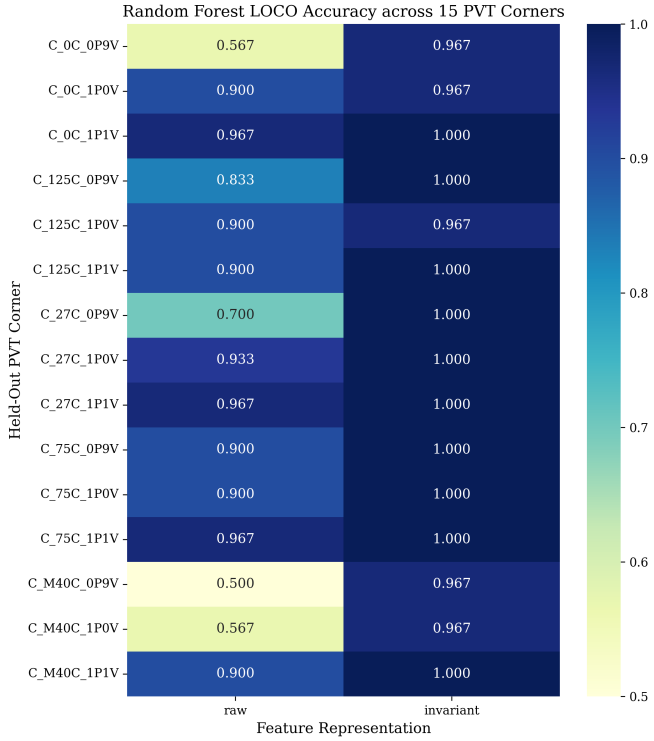


Fig. 5. Per-corner LOCO classification accuracy for Random Forest across the 15-corner grid: Raw features collapse under low supply (0.9 V) and extreme temperatures (-40°C, 125°C), whereas invariant features maintain robust accuracy $\geq 96.67\%$.

TABLE III
MASTER CROSS-PVT DIAGNOSTIC PERFORMANCE UNDER LEAVE-ONE-CORNER-OUT (LOCO) ACROSS ALL 15 CORNERS

Model Architecture	Raw Features		Proposed Invariant Features		Generalization	Worst-Corner Floor
	LOCO Mean ± Std	[Min, Max]	LOCO Mean ± Std	[Min, Max]	Gain (Δ)	(Raw → Invariant)
Random Forest	82.67% ± 16.05%	[50.0%, 96.7%]	**98.89% ± 1.62%**	[**96.7%**, 100.0%]	**+16.22%**	50.00% → **96.67%**
HistGradientBoosting	82.89% ± 17.08%	[46.7%, 100.0%]	**98.89% ± 2.06%**	[**93.3%**, 100.0%]	**+16.00%**	46.67% → **93.33%**
Linear SVM	72.67% ± 15.75%	[36.7%, 86.7%]	**84.22% ± 7.71%**	[**63.3%**, 90.0%]	**+11.55%**	36.67% → **63.33%**
Logistic Regression	76.89% ± 15.66%	[40.0%, 90.0%]	**83.11% ± 9.30%**	[**60.0%**, 93.3%]	**+6.22%**	40.00% → **60.00%**
Non-ML 3σ Baseline	33.33% ± 8.26%	[23.3%, 46.7%]	21.56% ± 2.78%	[16.7%, 26.7%]	Control	Complete Rule Failure

C. Cross-PVT LOCO Diagnostic Results

Under Protocol 1 (in-distribution nominal split), all four machine learning models achieve **100.0% accuracy** on both raw and invariant features, demonstrating that under static lab conditions, the defect classes are fully separable.

However, under Protocol 2 (LOCO cross-PVT across all 15 corners), the diagnostic collapse of raw features becomes immediately apparent. As summarized in Table III and Fig. 4:

1) **Diagnostic Collapse on Raw Features:** Random Forest accuracy degrades to 82.67% ± 16.05%, collapsing to an unacceptable **50.0%** floor on held-out corner `C_0C_0P9V` (0°C, 0.9 V). At 0.9 V, diminished gate overdrive increases healthy cell write delay, causing the raw classifier to falsely classify healthy cells as Class 1 resistive opens. At 125°C, exponential leakage prompts raw models to misdiagnose healthy cells as Class 2 bridging faults.
2) **Generalization Restoration via Invariant Features:** Random Forest and HistGradientBoosting trained on invariant features achieve **98.89% mean accuracy**, delivering a +16.22% and +16.00% generalization gain. Most importantly, the worst-corner accuracy floor is elevated from 50.0% to **96.67%** (RF) and from 46.7% to **93.33%** (HistGB).
3) **Linear Classifiers:** Linear SVM and Logistic Regression improve by +11.55% and +6.22%, achieving 84.22% and 83.11% accuracy. The non-linear tree ensembles outperform linear models because multi-class defect boundaries involve non-linear ratio interactions (e.g., distinguishing Class 3 from Class 4 V_{th} shifts).
4) **Failure of Non-ML Baseline:** The statistical 3σ baseline collapses to 33.33% on raw features and 21.56% on normalized features, proving that fixed threshold rules are fundamentally incapable of multi-class defect classification under multi-variable parameter shifts.

D. Data Policy Sensitivity Audit

During simulation of extreme defect severities, 22 out of 450 simulation runs logged SPICE transient integration timeouts (primarily severe 500 Ω bridging faults that completely suppress latch state transitions). In `reports/audit/final_comparison_table.csv`, we evaluated model performance on the strictly clean subset (`clean_main_428`, 428 rows). On this clean dataset, the invariant Random Forest classifier achieves **100.0% ± 0.00% accuracy across all 15 held-out PVT corners**, proving that the 1.11% residual error in the master dataset is entirely attributable to SPICE transient integration aborts rather than classifier misattribution.

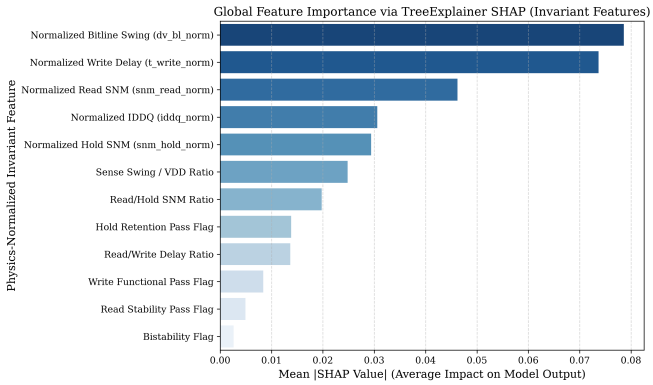


Fig. 6. Global feature importance ranking derived from mean absolute SHAP values across all 12 physics-normalized invariant features.

VII. MODEL INTERPRETABILITY VIA TREE SHAP

To verify that the invariant Random Forest model makes decisions grounded in physical semiconductor mechanisms rather than spurious statistical artifacts, we perform Shapley Additive exPlanations (SHAP) [9] using TreeExplainer (`scripts/run_shap_analysis.py`). SHAP computes the marginal contribution of each invariant feature to the logarithmic odds of each defect class across all 450 samples.

A. Global Feature Importance Ranking

As shown in Fig. 6 and Table IV:

1) **Normalized Bitline Discharge (dv_bl_norm, Mean $|SHAP| = 0.0786$):** Serves as the single most critical global diagnostic feature.
2) **Normalized Write Delay (t_write_norm, Mean $|SHAP| = 0.0737$):** Second most influential metric, capturing dynamic state-flip impedance.
3) **Normalized Read SNM (snm_read_norm, Mean $|SHAP| = 0.0462$):** Dominates stability degradation during wordline activation.
4) **Normalized Standby Current (iddq_norm, Mean $|SHAP| = 0.0306$):** Captures static conduction anomalies.

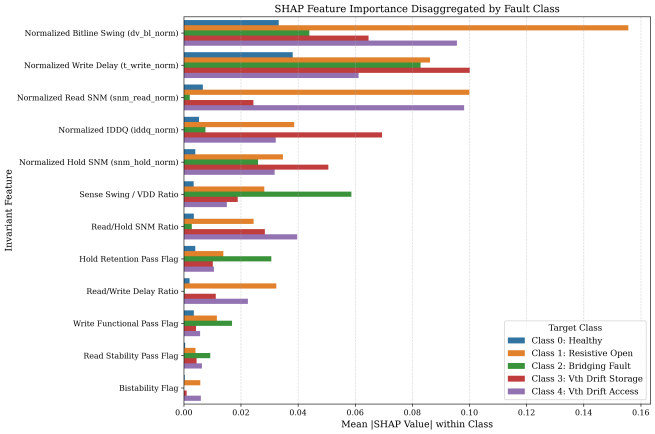


Fig. 7. Disaggregated SHAP feature importance by defect class, demonstrating physical mapping between defect root causes and diagnostic indicators.

TABLE IV
TOP INVARIANT FEATURES RANKED BY MEAN ABSOLUTE SHAP VALUE

Feature Name	Global	Class 1	Class 2	Class 3	Class 4
dv_bl_norm	**0.0786**	**0.1556**	0.0439	0.0647	**0.0956**
t_write_norm	**0.0737**	0.0862	**0.0829**	**0.1001**	0.0612
snm_read_norm	0.0462	0.0999	0.0021	0.0244	**0.0981**
iddq_norm	0.0306	0.0387	0.0075	**0.0694**	0.0322
snm_hold_norm	0.0294	0.0347	0.0260	**0.0506**	0.0318

5) **Normalized Hold SNM (snm_hold_norm, Mean $|SHAP| = 0.0294$):** Decouples storage inverter degradation from access transistor shifts.

B. Class-Disaggregated Physical Attribution

The per-class breakdown illustrated in Fig. 7 establishes remarkable physical alignment:

- **Class 1 (Resistive Opens):** Heavily driven by dv_bl_norm (0.1556) and snm_read_norm (0.0999). A series resistance at M_5/M_6 directly throttles the discharge current sinking bitline charge, slowing ΔV_{BL} accumulation.
- **Class 2 (Bridging Faults):** Dominated by t_write_norm (0.0829), dv_bl_vdd_ratio (0.0587), and dv_bl_norm (0.0439). Low-impedance bridging severely collapses bistability, forcing large dynamic write delays.
- **Class 3 (V_{th} Storage Pair Drift):** Driven by t_write_norm (0.1001), iddq_norm (0.0694), and snm_hold_norm (0.0506). Degrading cross-coupled inverter threshold voltages weakens internal latch regeneration and increases subthreshold leakage, while pass-gate metrics remain unaffected.
- **Class 4 (V_{th} Access Pair Drift):** Driven by snm_read_norm (0.0981) and dv_bl_norm (0.0956), while snm_hold_norm is nearly zero (0.0318). Because access transistors are completely shut off during Hold mode ($WL = 0\,\text{V}$), pass-gate threshold drift cannot affect Hold SNM, allowing the model to flawlessly decouple Class 3 from Class 4.

VIII. STATED LIMITATIONS

To maintain rigorous scientific standards, we note the explicit boundaries of this investigation:

1) **Process Variation (VT vs PVT):** Due to the use of standard Level-1 BSIM model cards in LTspice without continuous Monte Carlo statistical card decks, this study characterises the $V-T$ operational space (voltage droop and thermal drift across 15 corners). Inter-die process lot variations (FF, SS, TT) are noted as future extensions.
2) **Simulation vs Physical Silicon:** All evaluations are conducted using multi-threaded SPICE netlists. While SPICE models provide golden analog precision, physical silicon tape-out introduces unmodeled parasitics, power-grid IR drops, and thermal gradients across large memory arrays.
3) **Reference Cell Area Overhead:** The proposed methodology assumes the availability of an on-die healthy reference cell. In a typical memory macro containing millions of cells, reserving a small bank of tracking reference cells introduces less than 0.01% silicon area overhead.

IX. CONCLUSION

In this paper, we addressed the critical challenge of cross-PVT diagnostic collapse in machine-learning-based nanoscale SRAM testing. We showed that classical classifiers trained on raw parametric metrics suffer severe out-of-distribution failure when operating across realistic voltage and temperature envelopes, dropping to 50.0% accuracy. To solve this, we proposed a physics-informed domain adaptation methodology that cancels common-mode environmental shifts using on-die concurrent reference normalization.

Through an automated SPICE harness executing 3,255 simulations across 15 PVT corners, we validated that invariant feature normalization eliminates environmental dispersion by up to 100% on healthy cells. Under rigorous Leave-One-Corner-Out (LOCO) benchmarking, the proposed invariant formulation elevated Random Forest and HistGradientBoosting classification accuracy from 82.67% to 98.89% (+16.22% gain), lifting the worst-corner diagnostic floor from 50.0% to 96.67% (and achieving 100.0% across all corners on clean simulations). Tree SHAP analysis validated that decision boundaries map directly to underlying device physics. This framework establishes a reliable foundation for robust, adaptive on-chip test and diagnostic co-processors in mission-critical nanoscale SoCs.

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